

VeriScript - Project Close-out Report

On-chain smart contract source code verification for Cardano

Name of project and Project URL on IdeaScale/Fund

VeriScript - On-chain source code verification for Cardano

Project Catalyst URL:

<https://projectcatalyst.io/funds/14/cardano-use-cases-concepts/veriscript-on-chain-source-code-verification>

Your Project Number

1400122

Name of project manager

Adam Valach

Date project started

December 2025

Date project completed

May 2026

List of challenge KPIs and how the project addressed them

The proposal was funded under the *Cardano Use Cases: Concepts* challenge, whose KPIs centre on delivering a working, open-source product built on Cardano that solves a real ecosystem problem, demonstrates the EUTxO model in a novel way, and provides educational value and community engagement.

- **A working, open-source product built on Cardano.** VeriScript is a fully working platform live at <https://veriscript.org>, consisting of five Aiken smart contracts deployed on the Cardano preprod, a TypeScript/Node.js backend, a React frontend, and a Docker Compose environment for local reproducibility. All code is MIT-licensed and publicly available at <https://github.com/CardanoProphet/veriscript>.
- **A novel application of Cardano's EUTxO model.** VeriScript uses on-chain UTxOs as the canonical store of smart contract attestations. Each attestation is a UTxO at the `attestation_validator` address carrying the verified script hash, verifier identity (Signer token), and metadata - making

verification queryable by anyone reading the chain, with no centralized database or trusted intermediary.

- **A measurable improvement to ecosystem trust and security.** VeriScript provides the first decentralized, on-chain mechanism for Cardano to do what Etherscan's "Verified Contract" badge does for Ethereum - enabling users and wallets to confirm that a deployed address corresponds to the source code they have inspected. A reputation system and counter-attestation mechanism further incentivize accuracy and allow the community to dispute incorrect claims.
- **Open documentation and reproducibility.** Comprehensive technical documentation covering all five smart contracts, the datum/redeemer/parameter data types, and transaction diagrams is published in the repository. Build instructions for the Aiken contracts and Docker-based local setup let any developer reproduce the full environment from scratch.
- **Community engagement.** A public user test was conducted with 10 participants, achieving an average satisfaction score of 8.7/10 (87%), above the required 75% threshold. All reported bugs and improvement requests were triaged and resolved prior to close-out.

List of project KPIs and how the project addressed them

The project KPIs were defined by the four milestone outputs and acceptance criteria set out in the grant application.

- **M1 - Technical documentation.** Detailed documentation covering fundamentals, smart contract inner workings, the reputation system, datum/redeemer/parameter definitions, and transaction diagrams. Delivered as docs/veriscript.pdf. It covers all five validators and policies, the full UTxO flow for attestation, counter-attestation, and Signer token minting, with transaction diagrams for each core action.
- **M2 - Smart contracts, inline documentation for each function, and a "build" section in the documentation.** Five Aiken smart contracts delivered in contracts/validators/: attestation_validator, protocol_parameters, signature_token_policy, signer_metadata_validator, and signer_token_policy. Each function carries inline documentation. contracts/README.md contains the full build section, including a Nix-based reproducible toolchain and Docker alternatives. Contracts build without errors for both Mainnet and Preprod targets.
- **M3 - Backend, frontend, and local deployment.** The TypeScript/Node.js backend and React frontend are fully implemented and publicly available. docker-compose.yml, together with frontend/Dockerfile and backend/Dockerfile, brings both services up locally with a single command - enabling attestation, counter-attestation, and Signer token minting without additional setup.
- **M4a - Live deployment.** VeriScript is live at <https://veriscript.org> without performance issues or loading errors, with smart contracts deployed on the Cardano preprod.
- **M4b - Public user test: ≥10 participants, ≥75% average satisfaction.** 10 participants completed the test, with an average satisfaction score of 87% (requirement: ≥75%).
- **M4c - Bug resolution.** All reported issues were addressed: infinite wallet-loading state on reconnect (fixed with auto-reconnect logic), missing pointer cursor on buttons (fixed), truncated

transaction hashes in confirmation messages (replaced with block explorer links), inconsistent fonts in text fields (fixed), and a cost model error on Signer token minting (fixed).

- **M4d - Close-out report and video.** This document constitutes the close-out report; the close-out video is linked at the end.

Key achievements (in particular around collaboration and engagement)

- **Decentralized verification platform on Cardano.** VeriScript delivers what Cardano has lacked compared to Ethereum - an on-chain mechanism where any user can check whether the source code of a deployed contract has been independently verified. Five Aiken smart contracts are live on the Cardano preprod, handling attestation, counter-attestation, signer identity, and protocol governance.
- **End-to-end reputation and dispute system.** The platform implements a full verifier lifecycle: a developer mints a Signer token (establishing their on-chain identity), submits an attestation UTxO linking a script hash to source code, and any community member can submit a counter-attestation to dispute the claim. The attestation record is derived directly from on-chain state, with no off-chain trust required.
- **Reproducible local environment.** Any developer can clone the repository and bring up the full stack - backend, frontend, and contract integration - with a single `docker compose up`, without needing a live Cardano node for local development.
- **Comprehensive technical documentation.** `docs/veriscript.pdf` provides a detailed reference covering data types, transaction diagrams, and the rationale behind each design decision, intended to serve the wider Cardano developer community beyond VeriScript users alone.
- **Successful public user test.** 10 participants tested the live platform and provided structured feedback. The 87% average satisfaction score, combined with detailed bug reports, gave the team actionable signal to improve the product before close-out, and all reported bugs were resolved.
- **Open-source under MIT.** The full codebase - smart contracts, backend, frontend, deployment tooling, and documentation - is publicly licensed for re-use, modification, and incorporation into other projects.

Measurable impact at a glance:

- 5 Aiken smart contracts deployed on the Cardano Preprod testnet: *attestation_validator*, *protocol_parameters*, *signature_token_policy*, *signer_metadata_validator*, and *signer_token_policy*.
- 10 community members submitted structured feedback; average satisfaction 8.7/10 (87%), exceeding the 75% threshold.
- 7 bugs and improvement requests reported during user testing; all 7 resolved before close-out.
- Zero centralized trust required: all attestation records live as on-chain UTxOs, queryable by anyone, with no backend database as a single point of failure.
- Full reproducibility from a single `docker compose up`.

Key learnings

- **The EUTxO model is a natural fit for decentralized attestation.** Representing each attestation as a standalone UTxO - rather than rows in a database - made verification independently queryable and removed any single point of trust. This validated the core architectural bet and produced a reputation pattern (mint an identity token → produce attestation UTxOs → allow community dispute) that generalizes well beyond source-code verification.
- **On-chain data design has downstream UX consequences.** User testing surfaced that several validators require exactly one input, which wallets holding fragmented UTxOs cannot easily satisfy. Decisions made at the validator and datum level propagate directly into everyday wallet usability more than one may think.
- **Real users find issues that internal testing misses.** The public test revealed practical friction - an infinite wallet-loading state on reconnect, truncated transaction hashes, missing pointer cursors, inconsistent fonts, and a cost-model error on minting - that mattered far more to real users than to developers familiar with the system. Structured, scored feedback from outside the team proved to be the single most valuable quality signal of the project.
- **Reproducibility lowers the contribution barrier.** Investing early in a Nix toolchain and a one-command Docker Compose setup made the project demonstrably reproducible and significantly lowered the barrier for new contributors and reviewers to run the full stack.
- **Keeping all canonical state on-chain removes trust assumptions but shifts the work to reading the chain.** Storing every attestation as a UTxO rather than in a backend database means there is no central record to trust or to fail, but it also makes querying and indexing on-chain state the main engineering challenge rather than an afterthought. The trade-off was worth it for VeriScript, where independent verifiability is the entire point.

Next steps for the product or service developed

The funded scope - the smart contracts, the backend and frontend, the Docker environment, the user test, and the reports - is complete. Possible directions for follow-up work:

- **Deploy on mainnet.** Deploy the VeriScript protocol to Cardano mainnet, making attestations permanent and publicly verifiable on the live network.
- **Grow the active verifier network.** Engage existing Cardano developers, auditing firms, and tool builders to begin attesting well-known contracts. A critical mass of attestations is what makes the verification record meaningful to end users.
- **Integration with Cardano block explorers/wallets.** Surfacing VeriScript attestation status directly within explorers (e.g., Cardanoscan, Cexplorer) would make verification visible to the broadest possible audience without requiring users to visit veriscript.org separately.

Final thoughts/comments

VeriScript set out to close a concrete trust gap on Cardano: for most users, interacting with a smart contract means trusting the developer's word that a deployed address corresponds to the code they claim. The project delivered a complete, open-source, on-chain answer to that problem, demonstrating that the EUTxO model can serve as the foundation for decentralized, verifiable reputation rather than relying on centralized records.

We are grateful to Project Catalyst and the voting community for funding this work, and to the public testers whose feedback materially improved the product. We believe the most valuable outcome is not only the platform itself but the reusable on-chain attestation pattern it establishes - a building block we hope the wider Cardano community will adopt and extend. The team intends to continue developing VeriScript and welcomes collaboration from developers, auditors, and tooling providers interested in strengthening trust across the ecosystem.

Links to other relevant project sources or documents

- Public source repository: <https://github.com/CardanoProphet/veriscript>
- Smart contract source code: <https://github.com/CardanoProphet/veriscript/tree/main/contracts/validators>
- Technical documentation: <https://github.com/CardanoProphet/veriscript/blob/main/docs/veriscript.pdf>
- Backend source code: <https://github.com/CardanoProphet/veriscript/tree/main/backend>
- Frontend source code: <https://github.com/CardanoProphet/veriscript/tree/main/frontend>
- Docker Compose environment: <https://github.com/CardanoProphet/veriscript/blob/main/docker-compose.yml>
- Contract build instructions: <https://github.com/CardanoProphet/veriscript/blob/main/contracts/README.md>
- Live platform: <https://veriscript.org>

Link to Close-out video

<https://youtu.be/MaV05GDzEF8>